

Nikhita Anjani Ravi

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Computational Linguistics student and Undergraduate Researcher at Centre for Visual Information Technology, IIIT Hyderabad, with experience in NLP, CV, full-stack development, and machine learning for building scalable, secure, and production-ready systems.

Education

IIIT Hyderabad, B.Tech in Computer Science, MS in Computational Linguistics | **GPA:** 8.82/10 2024 - 2029

Experience

Undergraduate Researcher, Centre for Visual Information Technology, IIIT-H May 2026 - Present

- Advised by Dr. Ravi Kiran. Working on Computer Vision, Image understanding, Document parsing and applied ML for CV.

Software Developer Intern, Onicares Jan 2026 - Apr 2026

- Developed an automated OCR pipeline for medical reports, extracting key information into a structured format for maternal healthcare monitoring
- Contributed to a WearOS application for continuous tracking of heart rate, sleep patterns, SpO2, and other vital health metrics
- Improved battery efficiency, reduced load times, and streamlined handling of complex permission workflows for a smoother user experience

Full Stack Developer Intern, Morsebiz Jan 2026 - Mar 2026

- Architected and deployed two comprehensive web applications: a proforma invoice generation system with PDF and Excel export capabilities, and a complete invoicing platform
- Deployed product managers and buyer managers to handle thousands of rows of data at <0.5s speed and loading time.
- Eliminated manual operational bottlenecks through full-stack automation, significantly reducing processing time and improving data accuracy across the organization

Research Intern, Sambhasana Lab Dec 2025 - Present

- Working under Dr. Hrishikesh R. Terdalkar on knowledge graphs and LLM reasoning
- Recreated research papers and benchmarked results for Knowledge Graph creation pipelines
- Worked with RAG systems and data processing for 100K+ documents

Projects

Job Application Tracking System (MERN Stack) [Link](#)

- Engineered full-stack web application with React frontend and Express backend, implementing JWT authentication and protected routes to secure user data
- Designed RESTful API with MongoDB aggregation pipelines, enabling CRUD operations across 6 application stages with automated reminders and comprehensive interview tracking
- Integrated AI-powered resume scoring and advice/analysis system based on job descriptions and deployed real-time analytics dashboard with multiple interactive and engaging charts showing application statistics

AI-Powered Document Analysis System (RAG + LangChain) [Link](#)

- Built terminal-based quick RAG system processing PowerPoint and PDF documents with ChromaDB vector store, achieving 95% retrieval accuracy through optimized semantic chunking
- Integrated LangChain orchestration with multi-model LLM support (Llama, Mixtral via Groq API), enabling context-aware conversational AI with persistent memory and 3x faster inference
- Implemented efficient document parsing pipeline with batched embeddings and async processing, handling 1000+ page documents with sub-5s query response times

Custom Shell (C) [Link](#)

- Developed full-featured Unix shell from scratch with process management, I/O redirection, pipelines, and job control, handling 50+ concurrent background processes
- Optimized command parser using finite automaton-based tokenization, reducing parsing overhead by 35% and supporting complex command chaining

Technical Skills

Languages: JavaScript, Python, C, C++ , Java, SQL, Bash

Frontend: React, HTML/CSS, Material-UI, Responsive Design, REST API Integration

Backend & Database: Node.js, Express, MongoDB, JWT Authentication, WebSockets, RESTful APIs

AI/ML & Tools: LangChain, ChromaDB, Hugging Face, TensorFlow, scikit-learn, PyTorch, Git, Linux, Docker